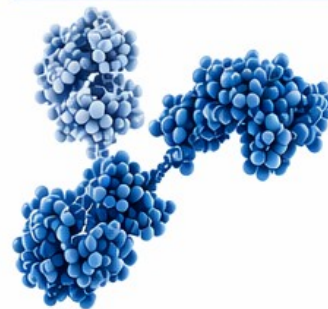


SIZE EXCLUSION CHROMATOGRAPHY (SEC)

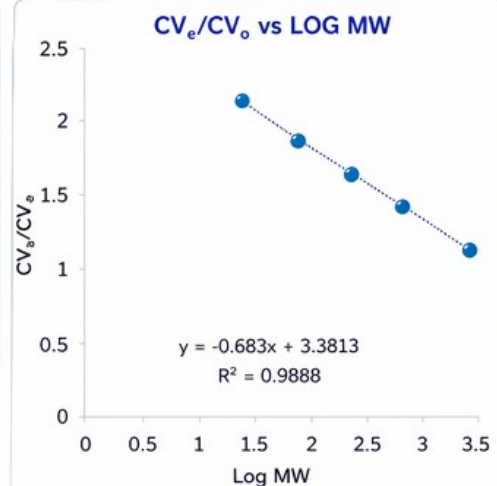
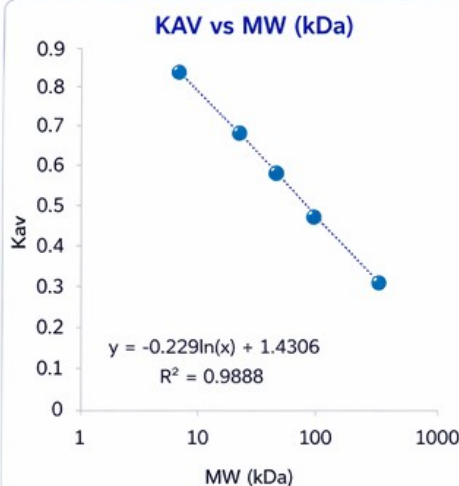
Molecular Weight Calibration & Column Performance



SEC analysis was performed to establish the relationship between K_{av} and molecular weight using a high-resolution agarose-based column and well-characterized protein standards.

METHOD SUMMARY

	Column Column Dia : 5 cm Column Length : 60 cm
	Bed Height 45 cm
	Column Volume ~ 884 mL
	Sample Volume 2 mL
	Flow Rate 4.42 mL/min (Linear velocity: 120 cm/h)
	Operating Pressure (ΔP) ~0.03 MPa
	System Suitability (NaCl Peak) Retention (CV) = 0.93 CV Area = 0.27 Height (mV) = 4.92 Plates per meter = 4057.71 HETP/Plate height = 0.02464 cm Asymmetry (T) = 0.89



High Linearity

Excellent linearity in both K_{av} vs MW and CV_e/CV_o vs $\log(MW)$ plots with $R^2 = 0.9888$



Reliable MW Estimation

Consistent and reproducible calibration ensures accurate molecular weight estimation within the working range.



Effective Fractionation Range

Validated fractionation range (globular proteins):
~15 kDa to 150 kDa
(Excludes V_o and V_t markers)

CALIBRATION STANDARDS AND RESULTS

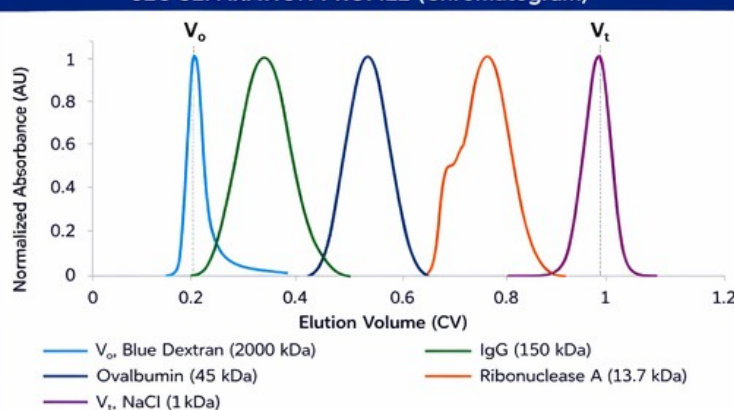
Protein / Standard	MW (kDa)	CV Peak (ml)	K_{av}
Blue Dextran (V_o Marker)	2000	334.59	—
Immunoglobulin (IgG)	150	413.68	0.33
BSA	66	468.27	0.43
Ovalbumin	45	556.29	0.57
Proteinase K	28.9	667.07	0.81
Ribonuclease A	13.7	715.61	0.85
NaCl (V_t Marker)	1	829.47	—

$$K_{av} = (V_e - V_o) / (V_t - V_o)$$

V_o = Void volume
 V_t = Total volume

CV_e = Elution volume of analyte
 V_o = Void volume (Blue Dextran)
 V_t = Total volume (NaCl)

SEC SEPARATION PROFILE (Chromatogram)



Chromatogram demonstrates size-dependent elution within the fractionation range (~15–150 kDa), with minimal peak overlap and consistent resolution.

System Suitability (NaCl Peak)

Peak	Retention (CV)	CV Area	Height (mV)	Plates per meter	Plate Height (HETP, cm)	Asymmetry (T)
1	0.93	0.27	4.92	4057.71	0.02464	0.89



High Resolution

Low plate height (0.02464 cm) and high plates per meter (4057.71) ensure superior resolution and peak separation.



Accurate MW Estimation

Excellent linearity ($R^2 = 0.9888$) provides reliable molecular weight estimation across ~15–150 kDa range.



Reproducible Performance

Consistent peak shape (Asymmetry = 0.89) and robust system suitability ensure reproducible results.



Broad Applicability

Ideal for desalting, buffer exchange, aggregate analysis and molecular weight determination of proteins and biotherapeutics.



Scalable & Reliable

Robust cross-linked agarose matrix delivers consistent performance across laboratory and process scales.

RESIN SPECIFICATIONS

- Resin Name : Protpure Agarose Precise
- Resin Type : Cross-linked Agarose 6% SEC Resin
- Pore Size : 32 – 45 nm
- Fractionation Range (globular proteins): ~15–150 kDa (Excludes V_o and V_t markers)
- pH Stability : 2 – 12 (working)
- Temperature Stability : 2 – 40 °C (working)
- Pressure Stability : Up to 0.3 MPa
- Chemical Stability : Compatible with common aqueous buffers and salts



APPLICATIONS

- Molecular weight determination
- Monomer & aggregate analysis
- Desalting and buffer exchange
- Purity and process development
- QC and lot release testing

ABOUT PROTPURE

ProtPure is an Indian bioprocessing company committed to delivering high quality, cost-effective chromatography solutions. Our SEC agarose resins are designed to meet global standards and support the success of your biologics journey.